LAB 01 | LINUX KERNEL BUILD AND BOOT

ARM Versatile Express • Linux 5.15.0 • BusyBox 1.35.0 • QEMU ARMv7
Evidence-based technical report | 20 October 2026

Verification scope. This report is based on configuration and startup scripts shared in the conversation and on the user-provided QEMU command results. It records an observed successful boot; it does not claim a new independent build or benchmark.

1. Objective and system overview

The goal is to bring up a small Linux userspace for ARM Cortex-A9 using an ARMv7-A Linux kernel, a Versatile Express Device Tree and a BusyBox-based compressed initramfs. The verified system runs within QEMU and exposes an interactive shell over the serial console.

Parameter	Verified or provided configuration
Emulated board / CPU	vexpress-a9 / ARM Cortex-A9 (ARMv7)
Kernel	Linux 5.15.0
BusyBox / filesystem	BusyBox 1.35.0 / initramfs
QEMU parameters	512 MiB RAM, 2 virtual CPUs, nographic serial console
Guest hostname	embedded-lab
Auxiliary bootloader build	U-Boot 2022.04 artifact retained; not part of tested direct boot

2. Submitted repository contents

The Lab01 repository package preserves the configuration snapshots, final boot artifacts and unpacked userspace. Upstream kernel/BusyBox/U-Boot trees and large compilation intermediates are excluded.

Location	Purpose
configs/kernel.config	Saved Linux kernel configuration
configs/busybox.config	Saved BusyBox configuration
configs/uboot.config	Saved U-Boot configuration
output/zImage	Booted ARM Linux kernel
output/vexpress-v2p-ca9.dtb	Versatile Express hardware description
output/initramfs.cpio.gz	Boot-tested compressed root filesystem
output/u-boot	Auxiliary U-Boot build artifact

3. QEMU boot configuration

The following command boots Linux directly: QEMU loads the kernel, DTB, and initramfs without executing U-Boot.

```
qemu-system-arm \
  -M vexpress-a9 -cpu cortex-a9 -m 512M -smp 2 \
  -nographic \
  -kernel output/zImage \
  -dtb output/vexpress-v2p-ca9.dtb \
  -initrd output/initramfs.cpio.gz \
  -append 'console=ttyAMA0,115200 rdinit=/sbin/init mem=512M'
```

Boot path: QEMU virtual hardware → kernel + DTB → initramfs root filesystem → BusyBox /sbin/init (PID 1) → /etc/inittab → /etc/init.d/rcS → interactive shell.

The kernel argument `rdinit=/sbin/init` directly selects BusyBox init. The separately stored top-level `/init` shell script is not the entry point for this QEMU run.

4. Userspace startup scripts

4.1 BusyBox inittab

```
::sysinit:/etc/init.d/rcS
::askfirst:-/bin/sh
::ctrlaltdel:/sbin/reboot
::shutdown:/sbin/swapoff -a
::shutdown:/bin/umount -a -r
::restart:/sbin/init
```

4.2 rcS boot script

```
#!/bin/sh
mount -t proc none /proc
mount -t sysfs none /sys
mount -t tmpfs none /tmp
echo 'Boot completed! Welcome to Embedded Linux Lab 1'
hostname embedded-lab
```

The script mounts the three pseudo-filesystems and establishes the hostname. The actual mount output below confirms that these mounts are present.

5. Verification evidence from QEMU

5.1 Kernel and CPU

```
uname -a
Linux embedded-lab 5.15.0 #1 SMP Mon Oct 5 16:09:28 +07 2026 armv7l GNU/Linux
```

The `/proc/cpuinfo` output reports two processor entries (0 and 1), ARMv7, CPU part 0xc09 and Hardware: ARM-Versatile Express. This is consistent with the requested Cortex-A9 configuration.

5.2 Filesystem mounts

```
rootfs on / type rootfs (rw,size=244912k,nr_inodes=61228)
none on /proc type proc (rw,relatime)
none on /sys type sysfs (rw,relatime)
none on /tmp type tmpfs (rw,relatime)
```

5.3 Init process and shell

```
/ # hostname
embedded-lab

/ # ps  (selected entries from the supplied output)
PID   USER     TIME  COMMAND
  1 root      0:02 init
 59 root      0:00 -/bin/sh
 64 root      0:00 ps
```

The student reached the `/ #` shell and ran `uname`, `cat /proc/cpuinfo`, `hostname`, `mount` and `ps` successfully.

6. Verification summary

Test item	Observed evidence	Assessment
Kernel boot	Linux 5.15.0 / armv7l	PASS
CPU enumeration	Processors 0 and 1, ARMv7, 0xc09	PASS
Board identity	ARM-Versatile Express	PASS
Init (PID 1)	BusyBox init in ps output	PASS
Root filesystem	rootfs mounted at /	PASS
Pseudo-filesystems	procfs, sysfs, tmpfs all mounted	PASS
Guest hostname	embedded-lab	PASS
Interactive shell	/ # prompt and executed commands	PASS

7. Limitations and reproducibility

The final initramfs archive is included, but Git cannot store Linux character or block device nodes as such. The Git-tracked rootfs tree excludes special entries such as /dev/console; they must be regenerated with appropriate device numbers and permissions when rebuilding a new archive. The verified archive in output/ should therefore be retained for direct reproduction of the shown QEMU boot.

The file output/u-boot and its configuration are preserved for completeness. The reported tests **do not validate a U-Boot boot path**, nor physical board deployment, kernel performance, network interfaces or comprehensive device-driver behavior. A clean rebuild would require the matching upstream sources and ARM cross-toolchain.

8. Conclusion

The observed QEMU session demonstrates a successfully initialized minimal Linux system for an ARMv7 Cortex-A9 target: the kernel reports version 5.15.0, BusyBox init runs as PID 1, standard pseudo-filesystems are mounted, and the serial shell is usable. The submitted configurations and boot artifacts provide a compact reference for future Embedded Linux labs.

Evidence source: QEMU terminal output supplied by the student in the conversation on 20 October 2026; previously supplied local build-directory filenames, initialization scripts, and configuration-file inventory.